Java Programming

WEEK 2 – Object-Oriented Programming (OOP Concepts)

Focus: Learn how to design reusable, scalable, and modular code using OOP principles.

Topics Covered:

- Classes and Objects
- Constructors and Overloading
- Inheritance and Polymorphism
- Abstraction and Interfaces
- Encapsulation and Access Modifiers

Assignments:

1. **Student Class Implementation**
    - Create a class with attributes and methods to display student info.
2. **Vehicle Inheritance Example**
    - Base class: Vehicle → Subclasses: Car, Bike (override methods).

Mini Project:

Library Management System (OOP)

- Classes: `Book`, `Library`, `User`
- Methods: Add Book, Borrow Book, Return Book
- Demonstrate inheritance, abstraction, and encapsulation

📽️ **Suggested Video:** OOP Concepts in Java Explained